
Page 1

In the name of God

University of Tehran

Faculty of Electrical and Computer Engineering

Course: Neural Networks and Deep Learning

Bonus Assignment

Question 1 ¹

- **Designer Teaching Assistant:** Tohid Abdi ²
- **Email:** tohid.abdi@ut.ac.ir ³

Question 2 ⁴

- **Designer Teaching Assistant:** Mohammad Mahdi Salmani ⁵
- **Email:** m.salmani78@ut.ac.ir ⁶
- **Submission Deadline:** 1404/04/01 (This corresponds to June 22, 2025) ⁷

Page 2

Table of Contents ⁸

- **Rules** ⁹
- **Question 1: Analysis of Neural Network Performance Under Adversarial Attacks** ¹⁰
 - 1-1 Training a ResNet Model on Noisy Images (25 points) ¹¹
 - 1-2 Transfer Learning with ViT on Flowers-102 (25 points) ¹²

- 1-3 Adversarial Attacks and Defense (30 points) ¹³
 - 1-4 Theoretical Questions (20 points) ¹⁴
 - **Question 2 - Generating Textual Descriptions for Images** ¹⁵
 - 2-1 Data Preparation (15 points) ¹⁶
 - 2-2 Implementing CNN-RNN Architecture with Attention Mechanism (25 points) ¹⁷
 - 2-3 Training and Evaluation (25 points) ¹⁸
 - 2-4 Analysis and Model Improvement (35 points) ¹⁹
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Page 3

Rules ²⁰

Please read the following carefully before answering the questions:

- Prepare a report of your answers in the format provided on the course page in the Elearn system, named REPORTS_TEMPLATE.docx. ²¹
- It is recommended to do the assignments in groups of two. More than two people are not allowed, and individual submissions will not receive extra points. Note that it is not necessary for group members to remain the same throughout the semester. ²²(i.e., you can do the first assignment with person A and the second assignment with person B, etc.). ²³
- The quality of your report is of special importance in the grading process. ²⁴Therefore, please mention all the points and assumptions you have made in your implementations and calculations in the report. ²⁵
- In your report, provide captions for figures and titles for tables as shown in the sample template. ²⁶
- It is not necessary to provide detailed explanations of the code in the report, but you must report and analyze the results obtained from it. ²⁷Analysis of the results is

mandatory, even if not explicitly mentioned in the question.²⁸

- Teaching assistants are not required to run your codes; therefore, bring any results or analysis requested in the question clearly and completely in the report. Failure to comply with this will result in a deduction from your assignment grade.²⁹
- Codes must be prepared in a notebook format with the .ipynb extension. At the end of the work, the entire code should be run, and the output of each cell must be saved in your submitted file. Therefore, if the output of a cell is a plot that you have included in the report, this plot must exist in both the report and the code notebook.³⁰
- In case of cheating, the score for all participants involved will be considered 100% [negative].³¹
- The only authorized programming language is Python.³²
- Using pre-written code for the assignments is strictly forbidden. If two groups use a common source and submit similar codes, it will be considered cheating.
- The late submission policy is as follows: After the submission deadline, you have up to one week to submit with a penalty. After this one week, your grade for that assignment will be zero.³³

Page 4

- First three days: No penalty³⁴
- Fourth day: 5% penalty³⁵
- Fifth day: 10% penalty³⁶
- Sixth day: 15% penalty³⁷
- Seventh day: 20% penalty³⁸
- The maximum score that can be obtained for each question is 100. If the total score for a question is more than 100, a score higher than 100 will not be applied.³⁹
 - For example, if the score obtained from question 1 is 105 and the score for question 2 is 95, the final assignment grade will be 97.5, not 100.⁴⁰

- Please place the report, codes, and other attachments in a folder with the following name, compress it, and then upload it to the Elearn system: ⁴¹

HW[Number]_[Lastname]_[StudentNumber]_[Lastname]_[StudentNumber].zip
(e.g., HW1_Ahmadi_810199101_Bagheri_810199102.zip)

For two-person groups, uploading the assignment by one member is sufficient, but it is recommended that both upload it. ⁴²

Page 5

Question 1: Analysis of Neural Network Performance Under Adversarial Attacks

This exercise is designed to integrate theoretical and practical concepts of deep learning, including sections on analyzing the effect of noise, transfer learning, adversarial attacks, and visual evaluation of models with explainability tools. ⁴³The theoretical part is also designed for a deeper understanding of the concepts. ⁴⁴

Important Note: In cases where the values of hyperparameters or exact settings are not specified, their selection is left to the student. In sections where the problem requires interpretation, using creativity and personal initiative will receive more points than asking direct questions. ⁴⁵

1-1 Training a ResNet Model on Noisy Images (25 points) ⁴⁶

In this section, you must train a ResNet model (such as ResNet18 or ResNet34) without using pre-trained weights (pretrained=False) on the CIFAR-100 dataset. ⁴⁷Then, investigate the effect of Gaussian noise on the model's performance. ⁴⁸

Steps:

1. Load the CIFAR-100 dataset. ⁴⁹
2. Add Gaussian noise with a mean of 0 and a variance of 0.005 to the training data. ⁵⁰
3. Train the ResNet model with the noisy data and then without noise, both for 20 epochs. ⁵¹
4. Plot the validation accuracy and loss graphs for both cases. ⁵²

5. Compare the performance of the models in both states and analyze the effect of the noise. ⁵³

1-2 Transfer Learning with ViT on Flowers-102 (25 points) ⁵⁴

In this section, transfer learning is performed on the Flowers-102 dataset using a pre-trained Vision Transformer (ViT) model. ⁵⁵

Page 6

Steps:

1. Load the vit_base_patch16_224 model with pre-trained weights and change the output layer of the model to classify 102 flower classes. ⁵⁶
2. Fine-tune the model for 5 epochs on Flowers-102. ⁵⁷
3. Now, load the same model without pre-trained weights and train it for 10 epochs. ⁵⁸
4. Plot the validation accuracy and loss graphs for both cases. ⁵⁹
5. By examining the details of the Flowers-102 dataset and the dataset on which the pre-trained model was fine-tuned, analyze the performance of the two models. ⁶⁰

1-3 Adversarial Attacks and Defense (30 points) ⁶¹

In this section, you will test and harden the models trained in the previous sections against FGSM and PGD adversarial attacks. ⁶²

Steps:

1. Implement the FGSM attack with a parameter of $\epsilon=0.1$ and the PGD attack with 7 iterations and a step size of $\alpha=0.02$. ⁶³
2. Generate adversarial images for the 4 models from the previous sections and evaluate their performance on these images. If you do not get the desired result, change the attack parameters. ⁶⁴

3. Retrain the models using adversarial training for 10 epochs. ⁶⁵
4. Compare the accuracy of the models in three states: without attack, under attack, and after defense. ⁶⁶

1-4 Theoretical Questions (20 points) ⁶⁷

In this section, you must answer conceptual questions descriptively and accurately. The use of formulas and explanatory diagrams is recommended. ⁶⁸

Questions: ⁶⁹

1. Although the ResNet model has millions of parameters, why does it still perform well in many cases with noisy data? ⁷⁰
2. Vision Transformer (ViT) models perform better when using pre-trained weights than without pre-training. Analyze this difference using the concepts of sharp and flat minima. ⁷¹
3. Analyze data augmentation as a form of adversarial training.
4. By analyzing the structure of ResNet and ViT and using the concepts of dropout and ensemble, justify the difference between these two models against adversarial attacks.
5. **[https://www.google.com/search?q=Optional Section \(5 points\):](https://www.google.com/search?q=Optional+Section+(5+points):)** For the ResNet and ViT models trained in sections 1-1 to 1-3, select samples from different classes and use Grad-CAM to show which areas of the image the model is focusing on. ⁷²

Page 8

Question 2 - Generating Textual Descriptions for Images ⁷³

Image Captioning is one of the important problems at the intersection of computer vision and natural language processing. This process usually involves extracting visual features from an image and converting them into descriptive sentences in natural language. For this purpose, various models are used, from the combination of convolutional neural networks (CNN) and classic language models like RNNs to more advanced architectures based on transformers

and machine vision, such as ViT. This technology is useful in applications such as making visual content accessible to the blind, image search, and content analysis on social networks.⁷⁴

2-1 Data Preparation (15 points)⁷⁵

For this exercise, we will use a translated version of the COCO Captions dataset, which includes 40,000 image-caption pairs. It is not necessary to use all the data for training.⁷⁶

Steps:

1. Get the data:

- Download the coco-flickr-fa-40k.zip data file from this link.⁷⁷

2. Pre-process images:

- Resize the images to a uniform size and normalize the pixel values to the [0, 1] range.⁷⁸

3. Pre-process captions:

- Normalize the Persian text data using the hazm library.⁷⁹ You can also remove punctuation, special symbols, and unnecessary numbers.⁸⁰

4. Create Vocabulary:

- Break the captions into individual tokens and create a dictionary of words. Assign a unique integer to each word.⁸¹
- Use special tokens like <eos>, <unk>, and <pad> appropriately.⁸²

5. Prepare for training:

- Divide the data into appropriate ratios for train, validation, and test sets.⁸³

* Pad the length of the captions using the '<pad>' token. Note that for this token, no error or gradient should be calculated during the training phase. [cite: 58]

6. Data Exploration:

- Display a few sample images along with their captions appropriately. ⁸⁴
- Report statistical analysis including a histogram of caption lengths, the 10 most frequent words, and the number of unique words in the vocabulary. ⁸⁵

2-2 Implementing CNN+RNN Architecture with Attention Mechanism (25 points)

The goal of this section is to implement an Image Captioning model using a combination of a Convolutional Neural Network (CNN) for extracting visual features and a Recurrent Neural Network (RNN) with an attention mechanism for generating captions.

1. Implement the Encoder:

- Choose a pre-trained CNN model like EfficientNet-B7.
- In this section, we need a matrix of vectors, where each vector represents the features of a region of the image. For this purpose, remove the final classification and pooling layers to extract the feature matrix. ⁸⁶

2. Implement the Attention Mechanism:

- Investigate what attention mechanisms were implemented in the "Show, Attend and Tell" paper.
- Implement the Additive Attention mechanism with the Soft Attention method.
 - Compare the current hidden state of the LSTM with the image features.
 - Calculate the attention weights.
 - Generate the context vector as a weighted average of the image features.

3. Implement the Decoder: ⁸⁷

- Use an Embedding layer to convert words into dense vectors. ⁸⁸
- [cite_start](#) You can use pre-trained embeddings like fastText for Persian. ⁸⁹
- *Reference paper: Show, Attend and Tell: Neural Image Caption Generation with Visual Attention* ⁹⁰

- * Use a single-layer or two-layer LSTM to generate the sequence of words. [cite: 65]
- * Feed the context vector generated by the attention mechanism into the LSTM at each time step. [cite: 66]
- * Use a Linear layer and a SoftMax function to predict the next word. [cite: 67]
- * Apply regularization techniques like dropout in the LSTM and weight decay in the optimizer. [cite: 68]
- * Report the total number of parameters in the model and the number of trainable parameters. [cite: 69]

2-3 Training and Evaluation (25 points) ⁹¹

1. Train the model:

- Use an appropriate loss function and optimizer and report your hyperparameters. ⁹²
- Use the teacher forcing method for training. ⁹³
- At the end of each epoch, display a sample image and its generated caption.
- Report the training and validation loss graph. ⁹⁴

2. Evaluate the model:

- Select a few random photos from the test dataset, run them through the model, and generate captions using Greedy Search and Beam Search (beam width = 3) methods.
- What metrics did the paper use for evaluation? ⁹⁵
- Evaluate the outputs with BLEU-1 and BLEU-4 metrics.
- For each of these images, display the real caption (reference text) and the generated captions. ⁹⁶

2-4 Analysis and Model Improvement (35 points)

1. Analysis of Model Errors:

- Select 5 sample images for which the generated captions are of low quality. ⁹⁷
- Plot the attention weights as a heatmap on the original image. ⁹⁸
- For each generated word, show a heatmap that specifies the areas the model focused on. ⁹⁹

- Analyze why the model made mistakes in these cases. For example, did it pay attention to the wrong areas of the image? ¹⁰⁰ Were inappropriate words chosen? Or is the problem related to the dataset or the architecture? ¹⁰¹

Page 11

2. Implement Scheduled Sampling: ¹⁰²

- In Teacher Forcing, the model always uses the reference captions for training. ¹⁰³ Scheduled Sampling uses the words generated by the model instead of the reference with a certain probability. ¹⁰⁴
- Implement a probabilistic mechanism that, at each training step, uses the reference word with probability p and the word generated by the model with probability $1-p$. ¹⁰⁵
- Decrease the probability p linearly or exponentially, for example, from 1 to 0.5 during training. ¹⁰⁶
- Add this mechanism to the model training code from the third section.
- Repeat the training and compare the error graph with the previous state. ¹⁰⁷
- *Reference paper: Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks* ¹⁰⁸

3. Experiment with Different Attention Mechanisms:

- The goal of this section is to implement the Scaled Dot-Product Attention mechanism instead of the method used in section 2-2. ¹⁰⁹ This attention mechanism, introduced in the Transformer architecture, uses a normalized dot product between the Query and Key vectors to calculate attention weights. ¹¹⁰
- Consider the current hidden state of the LSTM as the Query and the image features as the Key and Value.
- Calculate the dot product between the Query and Key and normalize it with a scaling factor of d_k , the dimension of the feature vector. ¹¹¹
- Use SoftMax to calculate the attention weights and then calculate the context

vector as a weighted average.¹¹²

- Repeat section 2-3 for this case and compare the results with the initial model.¹¹³